

Cost-aware reliability signal selection

Validate risk reduction before adding hidden-state or sampling cost

1. Controlled MCQ scoring

- Answerable benchmarks
- SLM option likelihoods
- Automatic correct/fail labels

2. Competing reliability signals

- Cheap: confidence, margin, entropy
- Hidden: summaries + PCA
- Sampling: 5× agreement

3. Selective-prediction validation

- AUROC failure ranking
- Risk at fixed coverage
- AURC, E-AURC, failure capture

4. Deployment decision rule

Start with confidence/option scores.
Add hidden probes only when stable
risk--coverage gains justify access,
latency, training, and maintenance.

Robustness layer

bootstrap CIs · option shuffle · five seeds · sample-size sensitivity · PCA dimension
layer localization · group permutation · latency / VRAM